

OSHADA NERANGA

Inventor | R&D Specialist | Sustainable Technology Innovator



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PROFILE

Award winning Innovator and R&D Specialist with a proven track record of developing sustainable, mechanical, and safety-driven technological solutions. Recognized at national and provincial levels, including multiple Gold and Bronze awards at the Ruhuna Invention & Innovation Exhibition. Key inventions span sustainable construction materials (upcycled porcelain alternatives), civic and emergency resource management (flood pre-warning systems and emergency water storage taps), and public safety hardware. Highly skilled in creative problem solving, rapid prototyping, and engineering sustainable solutions for real-world impact.

CORE SKILLS

- **R&D & Product Design:** Rapid Prototyping, Mechanical Mechanism Design, Product Development Lifecycle, Design for Manufacturing (DFM), Ergonomic Safety Tools, Reverse Engineering
- **Materials & Sustainable Tech:** Sustainable Composite Synthesis, Industrial Waste Upcycling (Porcelain Waste), Cold-Bonding Technology, Material Testing & Structural Characterization
- **Mechanical & Civic Systems:** Civic Pre-Warning & Sensor Alert Systems, Valve & Water Conservation Engineering, High-Voltage Spark Ignition Systems, Precision Tooling & Hardware
- **Intellectual Property & Commercialization:** Patent Application & Filing (LK, UK, AU - Patent Pending), Prior Art & Novelty Search, Patent Specification Drafting, Technical Exhibition Pitching
- **Software & Technical Tools:** 2D/3D CAD Modeling, Computer Literacy (OUSL Certification), Fabrication Equipment, Technical Documentation & Blueprinting

PATENTS

- **Sri Lanka Patent Application (Ref: LK/P/1/23387)** **Filed: December 13, 2024**
Title: Water Storing Building Block with Internal Reservoirs.
Status: Patent Application Filed
- **Sri Lanka Patent Application (Ref: LK/P/1/23235)** **Filed: May 22, 2024**
Title: A Bulb Changing Device for Replacing Bulbs Fitted in Permanently Mounted Holders or Holders Attached to End of a Wire.
Status: Patent Application Filed
- **Australia Patent Application (Ref: 2026905817)** **Filed: June 06, 2026**
Title: Eco-Friendly, High Strength Construction Brick Manufactured from Waste Porcelain Using High Pressure Cold-Bonding Technology with a Ball Clay Catalyst and Method of Production Thereof.
Status: Patent Application Filed
- **United Kingdom Patent Application (Ref: GB2613415.5)** **Filed: June 06, 2026**
Title: Eco-Friendly, High Strength Construction Brick Manufactured from Waste Porcelain Using High Pressure Cold-Bonding Technology with a Ball Clay Catalyst and Method of Production Thereof.
Status: Patent Application Filed

KEY INVENTIONS & R&D PROJECTS

Eco-Friendly Porcelain Construction Brick (Sand & Shingle Replacement)

- Formulated an eco-friendly, high-strength structural brick utilizing recycled industrial porcelain waste and cold-bonding technology, significantly reducing reliance on natural sand and fired clay.
- Filed patent applications across Sri Lanka, the United Kingdom, and Australia (Patent Pending)
- Honored with Gold Award (1st Place - Provincial) at Ruhuna Exhibition (2024) and Provincial Level Certificate at Sahasak Nimavum (2024), selected for further technical modifications.

[\[Gold Award - RIIIE\]](#) | [\[Certificate - Shasak Nimavum\]](#)

Pre-Warning Flood Detection & Civic Early Alert System

- Engineered a civic-grade flood detection mechanism designed to provide automated pre-warning alerts in flood-prone geographical zones.
- Honored with 3rd Place at the Entrepreneurship Skills Exhibition & Competition (2025) and selected for further technical modifications.

[\[3rd Place - Entrepreneurship Skills Competition\]](#)

Emergency Water Stock & Conservation Tap

- Developed a specialized mechanical valve tap system designed for crisis water rationing and municipal conservation during resource shortages.
- Awarded Bronze Award (3rd Place - Provincial) at Ruhuna Exhibition (2024) and Provincial Level Certificate at Sahasak Nimavum Exhibition (2024), selected for further technical modifications.

[\[Bronze Award - RIIIE\]](#) | [\[Certificate - Shasak Nimavum\]](#)

Night-Visible High-Contrast Vehicle Plate

- Formulated a passive visibility number plate system for enhanced road safety.
- Awarded National Level Certificate at the Road Safety Invention & Innovation Challenge (2024) and selected for further technical modifications.

[\[Certificate - Road Safety Invention & Innovation Challenge\]](#)

Universal Overhead Bulb Changing Mechanism

- Designed an ergonomic mechanical device capable of safely replacing light bulbs fitted in permanently mounted high sockets or wire-pendant suspended holders without secondary elevation tools.

Luminous Emergency Torch & High-Voltage Sparker

- Fabricated a high efficiency luminous emergency torch and a high-voltage spark ignition safety mechanism.

[\[Luminous Torch - RIIIE\]](#) | [\[Luminous Torch - Sahasak Nimavum\]](#) | [\[High-Voltage Sparker - Sahasak Nimavum\]](#)

Mechanical Hardware & Maintenance Tools

- **Interlock Pavement Gap Grass Cutter:** Designed a specialized cutter to clear weeds between interlock pavement gaps.
- Honored with Best Emerging Inventors Award at Ruhuna Innovation & Invention Exhibition (2023).

[\[Best Emerging Inventor - RIIIE\]](#) | [\[Certificate - RIIIE\]](#)

- **Wide Angle Rotational Camera:** Engineered a rotational camera mount to expand field of view and eliminate surveillance blind spots.

[\[Certificate - RIIIE\]](#)

EDUCATION

Computer Literacy Certificate Course

2025 - Present

The Open University of Sri Lanka

Primary & Secondary Education

2014 - 2024

G/ Udugama National College, Galle

REFERENCES

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